

**SPL1 Project Proposal Form, 2025**  
**Institute of Information Technology (IIT)**  
**University of Dhaka**

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| <b>Student's Roll:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | M. Hasebur Rahman Sinha |               |  |
| <b>Student's Roll:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1620                    | <b>Phone:</b> |  |
| <p><b>Project Description:</b></p> <p><b>“Exploring Different Modes of Operation of Block Ciphers”</b></p> <p><b>1. Modes of operation:</b></p> <p>In symmetric block ciphers (AES, DES, 3DES) the algorithm works on a fixed amount of bits to produce ciphertexts. They require modes of operation to properly encrypt and decrypt real life data.</p> <p><b>2. Problem statement :</b></p> <p>While modern block ciphers are independently mathematically secure, improper use of them can give away vulnerabilities or create inefficiencies. Choosing an appropriate mode is critical for maintaining both security and performance of encryption systems. I aim to compare and analyze the usage of these modes of operation of different encryption techniques to show each of their strengths, weaknesses and appropriate use.</p> <p><b>3. Objectives:</b></p> <p>Implement NIST standardized mode of operations</p> <ol style="list-style-type: none"><li>1. ECB (Electronic Cookbook)</li><li>2. Cipher Block Chaining (CBC)</li><li>3. Cipher Feedback (CFB)</li><li>4. Output Feedback (OFB)</li></ol> <p>Implementation of math functions, pseudo random number generator for initial vector, padding schemes for bit management and other functions required. Analysis report compared with papers of NIST and IEEE.</p> |                         |               |  |

Block Ciphers used to test these:

1. DES
2. 3DES
3. AES

The project will be entirely terminal based.

**Languages or Tools to be used:**

C, GCC/G++ , git for version control, standard libraries of C etc

**Supervisor's Name:** Dr. Md. Shariful Islam \_\_\_\_\_

**Signature of the supervisor:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Before the Midterm Presentation:**

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

**Signature of the supervisor:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Midterm Presentation Feedback:**